

LIAWENEE, Australia

WMO: 959590

Lat: 41.8997S

Lon: 146.6694E

Elev: 3465

StdP: 12.95

Time Zone: 10.00 (AUS)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.7	24.3	17.1	14.8	39.3	21.0	17.9	37.4	33.7	33.2	29.7	34.0	2.4	270	0.768

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.8	75.9	56.9	72.1	55.5	68.9	54.5	60.2	69.9	58.4	67.1	56.7	65.0	14.6	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.1	79.2	61.8	55.2	73.6	60.7	53.4	68.9	59.1	28.1	70.1	26.8	66.9	25.6	65.3	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.9	25.5	23.0	16.5	82.1	1.9	3.8	15.2	84.9	14.1	87.1	13.0	89.3	11.7	92.1
			15.2	64.4	2.1	1.8	13.7	65.7	12.4	66.8	11.2	67.9	9.7	69.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.6	52.8	52.1	49.2	44.0	39.9	36.6	35.3	35.9	38.7	42.5	47.0	49.5
	DBStd	8.23	7.19	5.94	6.49	5.30	4.73	4.08	3.72	4.29	4.47	5.94	6.38	6.15
	HDD50	2785	48	41	94	191	313	401	457	438	340	246	132	84
	HDD65	7828	386	362	492	629	777	851	922	903	789	697	539	481
	CDD50	445	136	101	69	12	1	0	0	0	1	13	43	69
	CDD65	10	8	1	1	0	0	0	0	0	0	0	0	0
	CDH74	196	124	20	23	0	0	0	0	0	0	0	5	24
	CDH80	26	21	3	1	0	0	0	0	0	0	0	0	1
(14) Wind	WSAvg	11.4	11.9	10.5	10.2	10.4	11.5	10.3	11.3	11.8	12.5	12.3	12.1	12.1
(15) Precipitation	PrecAvg	47.6	3.0	2.3	2.8	3.3	4.2	4.8	5.9	6.3	5.2	4.0	3.3	3.1
	PrecMax	61.5	7.2	5.9	6.5	6.9	10.0	10.2	8.6	11.7	9.0	7.4	6.3	6.6
	PrecMin	31.8	0.5	0.4	1.1	1.3	1.3	0.9	3.3	1.8	2.0	1.4	0.6	1.0
	PrecStd	5.7	1.8	1.2	1.5	1.4	2.4	1.9	1.6	2.5	1.9	1.4	1.6	1.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	83.0	76.4	77.0	65.5	58.4	51.6	47.3	53.5	58.4	69.0	73.7	78.1
		MCWB	60.8	57.9	57.0	53.3	46.4	43.7	42.8	42.3	46.2	51.3	54.1	57.8
	2%	DB	77.5	72.3	70.9	60.7	53.0	47.8	44.6	48.8	53.9	63.6	69.3	71.6
		MCWB	57.7	56.3	55.5	50.4	45.8	43.8	41.0	42.4	44.4	48.0	53.0	54.4
	5%	DB	72.7	68.9	66.4	57.3	50.1	45.5	43.0	45.7	50.3	58.9	65.3	67.2
		MCWB	56.2	55.7	54.3	48.9	45.5	42.7	40.1	41.1	44.0	47.4	51.5	53.3
	10%	DB	68.8	65.3	62.4	53.9	48.1	43.7	41.4	43.5	47.4	54.9	61.2	63.3
		MCWB	54.6	54.3	52.7	48.2	44.8	41.4	39.2	40.4	42.6	46.3	50.7	51.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.1	61.7	60.8	54.9	50.6	47.9	44.8	45.6	49.8	54.8	58.3	60.8
		MCDB	76.9	70.0	69.6	60.6	53.8	49.1	45.9	49.0	53.3	61.0	67.5	69.4
	2%	WB	60.6	59.4	58.3	52.7	48.1	45.1	42.4	43.8	47.0	52.2	55.7	57.8
		MCDB	72.0	67.6	65.8	57.1	50.5	46.6	44.0	47.3	51.3	58.8	64.4	66.2
	5%	WB	58.4	57.7	56.3	51.0	46.7	43.5	41.1	42.4	45.2	49.7	53.9	55.5
		MCDB	68.7	65.2	63.6	55.6	49.2	45.4	42.6	45.4	49.5	56.2	62.2	63.3
	10%	WB	56.1	55.9	54.3	49.1	45.3	42.0	39.9	41.0	43.6	47.5	51.7	53.4
		MCDB	65.8	62.9	61.1	53.6	47.9	43.8	41.5	43.7	47.7	54.1	59.4	61.9
(35) Mean Daily Temperature Range	5% DB	MDBR	22.8	21.0	20.3	16.3	13.3	11.9	10.9	12.3	15.1	18.7	20.1	20.7
		MCDBR	29.6	27.2	27.2	23.4	17.4	16.1	13.4	17.3	20.6	26.8	27.6	28.2
	5% WB	MCWBR	14.9	14.6	15.1	13.5	11.1	10.8	10.1	11.1	12.9	14.9	15.1	15.6
		MCDWR	25.3	22.0	22.3	17.4	13.2	12.2	10.9	12.9	16.9	21.1	23.3	23.3
(40) Clear-Sky Solar Irradiance	taub		0.302	0.295	0.287	0.278	0.266	0.257	0.259	0.272	0.289	0.293	0.299	0.296
		taud	2.589	2.625	2.645	2.641	2.652	2.663	2.664	2.630	2.566	2.569	2.572	2.600
	Ebn at Noon		326	320	310	291	274	267	274	289	303	317	325	330
		Edh at Noon	32	30	26	23	19	17	18	22	28	31	33	32
(44) All-Sky Solar Radiation	RadAvg		2153	1832	1380	933	586	458	507	753	1115	1593	1926	2107
	RadStd		154	88	67	59	42	42	27	60	76	93	148	142

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (36 neighbors)	+0.58	N/A	N/A	+1.21	+0.59	N/A	N/A	-176	+191	+22

Nomenclature: See separate page